

CSV Data Quality Triage Agent

Lab 9 - Activity 2 | Phase 3 technical report | 2 October 2026 | LLM synthesis revision

1. Problem and implemented workflow

CSV defects can distort machine-learning preparation. The application answers a user's diagnostic question using selected deterministic checks, with evidence and contextual recommendations. It does not clean datasets or train models. The local release upload limit is 250 MiB (MiB = 1,048,576 bytes); supported shapes and measured boundaries are reported on page 4.

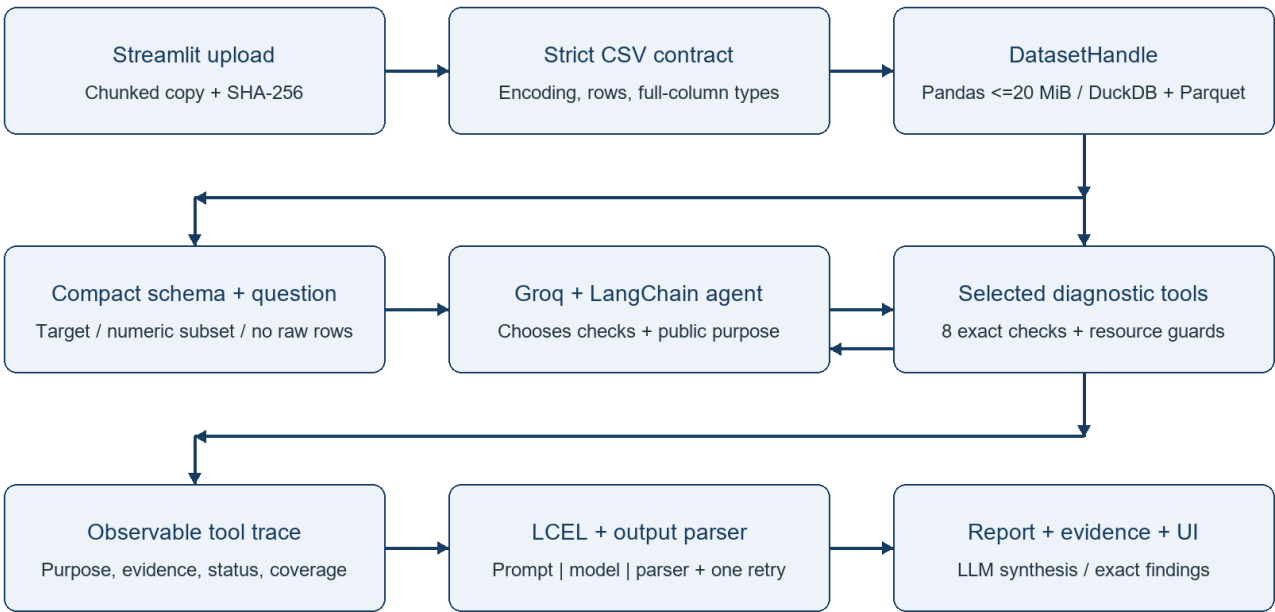


Figure 1. Local dataset access, selective LLM-tool interaction, and evidence-validated reporting.

LangChain components and external API

Component	Implemented role
ChatPromptTemplate	Agent instructions include question, schema, target, allowed tools and budgets.
Agent + StructuredTool	create_agent asks Groq openai/gpt-oss-120b to select from eight checks.
LCEL chain	REPORT_PROMPT model StrOutputParser produces candidate report JSON.
PydanticOutputParser	Parses LLM summary, cited interpretations and proposed next steps; exact issue validation plus narrative guards; one repair attempt.

The Groq API uses langchain-groq and an ignored root .env key. Raw datasets and row previews stay local. Schema, questions, aggregate evidence and class labels can reach the provider and may be sensitive. An optional brief public reason explains each check before execution and appears in progress/trace; private chain of thought is not recorded.

2. Implementation and exact diagnostics

The uploader is copied and hashed in chunks once. Strict validation handles UTF-8/BOM and Windows-1252, rejects inconsistent fields and empty headers, numbers duplicate names, preserves leading-zero identifiers, and examines complete columns before type selection. Both engines share this contract. Small files use the Pandas reference; larger files use DuckDB and typed Parquet in an app-owned OS temporary directory.

Diagnostic	Evidence and correctness rule
Profile	Rows/types/nulls and exact distinct counts; at most 50 visible columns with the selected target retained.
Missing	Exact null counts and percentages; existing 5%/30% severity thresholds.
Duplicates	Rows minus distinct complete rows; nulls equal; context required before removal.
Constants	Distinct categories include null; near-constant dominant fraction at least 95%.
Cardinality	Eligible categorical or identifier-named columns; at least 20 observations and 90% unique ratio.
Outliers	Finite numeric values, exact Type-7 quartiles and 1.5 x IQR bounds; at least four observations.
Imbalance	Selected target; exact class counts, rounded percentages and unchanged ratio 3/9 thresholds; unsuitable/long-label targets skipped.
Correlation	Exact finite pairwise Pearson r; target excluded; r at least 0.95; does not establish leakage.

Execution controls and evidence integrity

An owned background job admits one heavy operation. DuckDB uses a 1 GB memory setting, two threads and 2 GiB spill allowance; the session storage quota is 4 GiB. Query cancellation interrupts the connection and waits for active work before cleanup. Reset, replacement and failed import close handles; abandoned storage cleanup checks ownership. Memory settings do not guarantee whole-process RSS.

Numeric scans require at most 20 selected columns; full-row duplicates are guarded above 50 columns. Exact quartiles use one shared aggregate state plus a conservative allocation check. A resource-limited operation is skipped or fails visibly; it never silently samples. Six executed checks, twelve attempted events, twenty emitted findings and a 16 KiB observation cap bound orchestration. Full local evidence is separately downloadable.

Structured issues must exactly match transmitted successful findings; missing supported issues are restored. The LLM-written summary, cited interpretation and proposed next steps are retained. `tools_used`, `limitations` and `assessment_summary` are derived from actual events and coverage. Caches include dataset, scope and thresholds; cache hits are disclosed.

Interpretation citations must name called tools. Narrative numbers must occur in observed results, restricted to cited tools for interpretation; broad clean/safe/leakage assurances are rejected. These guards do not prove correct number-to-column attachment, causality or semantic accuracy. Qualitative interpretation and suggestions remain model output requiring review; exact issue matching is a separate stronger guarantee.

3. Evaluation and failure handling

The original 1 October 2026 baseline had 122 passing tests and the live results below. Its PDF/MD/results are archived in [submission/archive/phase3_original_122_tests/](#). Tests cover backend parity, adversarial CSV values, actual interruption, cancellation, cleanup/cache and both handles through the LangChain graph. These original live and scaling records remain historical evidence; no new 250 MiB benchmark was run for the synthesis revision.

ID / case	Latest result	Actual evidence / elapsed
T01 Missing values	pass	missing_values_check; 3.09 s
T02 Broad quality	pass	missing_values_check, constant_columns_check, duplicate_rows_check; 52.43 s
T03 Target imbalance	pass	class_imbalance_check; 12.53 s
T04 Outlier/correlation	pass	outlier_check, correlation_check; 38.40 s
T05 Unsuitable target	pass	class_imbalance_check; 4.33 s
T06 Injected tool failure	pass	duplicate_rows_check, duplicate_rows_check, duplicate_rows_check, duplicate_rows_check; 7.69 s
T07 No target selected	pass	No tool calls; 11.16 s
T08 Clean missing check	pass	missing_values_check; 15.96 s
T09 Header-only upload	pass	No tool calls; 0.02 s

LLM synthesis revision: fresh verification

The revised suite has 149 passing tests; its separate record is [docs/evaluation/phase3/llm_synthesis/test_results.txt](#). Fresh live cases below verify model-written synthesis, cited interpretation, deterministic assessment and visible public selection reasons. Runtime success and numeric/citation guards do not establish full semantic proof.

Fresh case	Result	Calls / elapsed
N01 Moderate 60/40	pass	class_imbalance_check; 5.12 s
T04 Numeric features	pass	outlier_check, correlation_check; 27.78 s
T07 Missing target	pass	No tool calls; 2.76 s

N01 explains the observed 60/40 distribution and 1.5 ratio with no flagged issue, using unchanged 3/9 thresholds. T07 provides target-selection guidance without a diagnostic call. Its first revision attempt failed with HTTP 429; that record remains alongside the successful retry. Public selection reasons appear only when a check is called.

A separate fresh browser run of the moderate target case took 4.6215s. Its downloaded [ui_evidence.json](#) and screenshots [11_llm_summary](#), [12_llm_interpretation](#) and [13_llm_next_steps](#) show the public selection reason, LLM synthesis, verified scope and suggested actions.

Observed challenge and controlled failure

Original attempts retain triage errors, repeated checks and Groq HTTP 429. T06 injects an exception only into the duplicate diagnostic: its trace records the error and a limitation, with no invented count. T05 uses an unsuitable `record_id` target. T09 rejects a header-only upload before any provider request despite the harness live-mode label. Unsupported issues are rejected after one repair retry; selected coverage remains visible.

Original evidence: [docs/evaluation/phase3/scenarios/*_result.json](#) plus companion [trace/report JSON](#) and [scenario_results.md](#). Revision evidence: [docs/evaluation/phase3/llm_synthesis/](#). Raw records retain hashes, prompts, target, provider/model, actual calls and latency. Scripted tests establish wiring and guard behavior; live records establish observed provider behavior.

4. Measured scaling, limits and submission

The generator provides independent construction-based null, duplicate and class-count manifests. Latest attempts per profile/size/repeat show slowest import and peak RSS/storage below. Pass means verified counts/shape and resource gates, not independent verification of every large-file statistic. Guarded means expected skips with those statistics unassessed. OS caching is uncontrolled. The diagnostic harness excludes browser/uploader and API calls.

Profile / MiB	Runs	Max import s	Peak RSS MiB	Peak temp MiB	Gate
tall numeric / 1	3	0.27	110.0	2.0	pass
tall numeric / 20	3	3.52	158.8	42.9	pass
tall numeric / 50	3	11.07	240.7	105.4	pass
tall numeric / 100	3	21.47	353.2	214.8	pass
high unique strings / 250	3	36.61	730.2	506.0	pass
null heavy / 250	3	96.98	1076.0	574.9	pass
quoted unicode / 250	3	23.62	1002.4	528.0	pass
tall numeric / 250	3	62.09	813.0	534.9	pass
wide numeric / 250	3	78.98	959.2	564.9	guarded

Across the latest successful 250 MiB profiles, slowest measured selected-check times were class imbalance 0.15s; constant columns 1.33s; correlation 8.01s; dataset profile 1.22s; duplicate rows 2.06s; high cardinality 0.45s; missing values 0.00s; outlier 3.29s. These are diagnostic times; provider/report latency is measured separately in the live scenario table.

All eight diagnostics completed in the saved 250 MiB risk runs for high unique strings, tall numeric. Large-file constants, cardinality, outliers and correlation lack independent full-size statistical ground truth; successful execution is distinct from measured statistical correctness. Small-fixture backend parity provides separate evidence.

The initial 60-second full-validation goal failed on 250 MiB tall data; those failed attempts are preserved. The release import budget was revised to 120 seconds after observing the cost of strict full-column validation. Correctness, 2 GiB application/worker RSS and other declared gates are still required; changing the time budget is explicitly documented, not a claim that the initial goal passed.

Actual browser upload: 262,143,987 bytes, 3,274,752 rows and 13 columns loaded into DuckDB in 36.093s, excluding HTTP transfer. Sampled peak app/worker RSS was 1312.97 MiB including the Streamlit upload buffer; browser RSS was excluded. A live Groq missing-value run took 3.9375s. This is one desktop upload, not a concurrency test. Original downloaded evidence and resource record are in docs/evaluation/phase3/ui/.

Streamlit UploadedFile retains an in-memory buffer. Disk-backed analysis avoids full DataFrames and extra complete byte copies but is not an end-to-end streaming upload service. The demonstrated envelope concerns tested shapes, not every possible 250 MiB CSV. Wider schemas, huge labels, holistic quartile allocations, disk pressure and provider rate limits can lead to explicit skips/errors. Outliers, duplicates and correlations require domain interpretation.

Reproducibility and remaining human deliverables

Reproduce using README commands, saved environment/source hashes, and scripts/build_phase3_report.py with the verified test count. The silent two-case demo is 6:30 with a matched script; contributions follow the confirmed split. Add narration, sign statements, complete the real feedback form and confirm course instructions. The brief's deadline was 30 September 2026. See submission/README.md.

Sources

Assignment: docs/source/Lab 9_2026_27.docx (Activity 2, Phase 3 and rubric); Phase 1 proposal in docs/source.
Technical references: docs.streamlit.io/develop/api-reference/widgets/st.file_uploader;
duckdb.org/docs/current/guides/performance/how_to_tune_workloads;
duckdb.org/docs/current/sql/functions/aggregates. Implementation, saved traces and benchmark manifests are the sources for measured claims.